

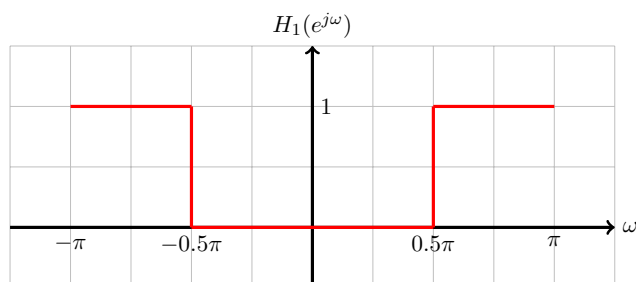
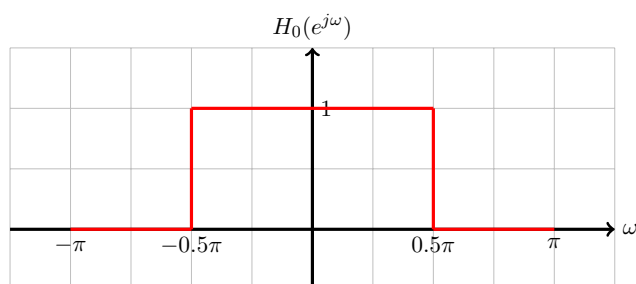
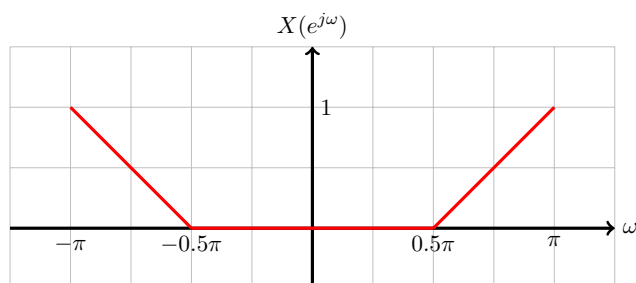
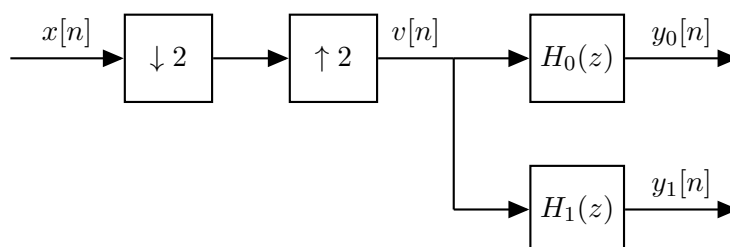
Homework #7

Due: 11:59pm, November 25

Objectives: Study polyphase decomposition and multirate systems

1. [Vaidyanathan 4.1]

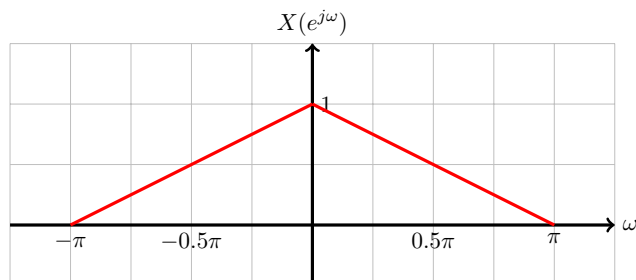
Consider the system in the following block diagram with input transforms and responses as indicated.



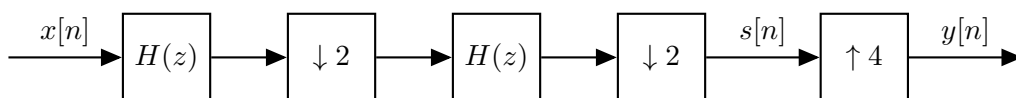
Sketch $Y_0(e^{j\omega})$ and $Y_1(e^{j\omega})$.

2. [Vaidyanathan 4.11]

Consider a sequence $x[n]$ with $X(e^{j\omega})$ as shown in the following figure.



Suppose we generate the sequences $y(n)$ and $s(n)$ from $x(n)$ by the following system



where

$$H(e^{j\omega}) = \begin{cases} 1 & |\omega| < \pi/2 \\ 0 & \pi/2 \leq |\omega| \leq \pi. \end{cases}$$

Plot $S(e^{j\omega})$ and $Y(e^{j\omega})$.

3. [Vaidyanathan 4.12]

Let $H(z)$ represent an FIR filter of length 10 with impulse response coefficients $h[n] = (1/2)^n$ for $0 \leq n \leq 9$ and zero otherwise. Find the polyphase components $H_0(z)$ and $H_1(z)$.

4. [Vaidyanathan 4.13]

Let $H(z) = (a + z^{-1})/(1 + az^{-1})$.

- Compute its polyphase components $H_0(z)$ and $H_1(z)$ so that $H(z) = H_0(z^2) + z^{-1}H_1(z^2)$.
- Compute the magnitude response of $H(z)$. Is $H(z)$ an allpass system?
- Compute the magnitude response of $H_0(z)$ and $H_1(z)$. Are $H_0(z)$ and $H_1(z)$ also allpass systems?